

Question 13 – Oscilloscope and A.C. Signals

(a) State what is meant by the term alternating current (AC)

✓ 1 mark

✓ **Answer:**

An alternating current (AC) is an electric current that **regularly reverses direction**.

(b) Determine the RMS voltage of the signal

✓ 3 marks

- From the oscilloscope trace:
 - Y-gain** = 1.0 V/div
 - Peak-to-peak height = 4.0 divisions
 - So, **peak voltage** $V_{\text{peak}} = \frac{4.0}{2} \times 1.0 = 2.0 \text{ V}$

$$V_{\text{rms}} = \frac{V_{\text{peak}}}{\sqrt{2}} = \frac{2.0}{\sqrt{2}} = 1.41 \text{ V}$$

(c) Determine the timebase setting on the oscilloscope

✓ 3 marks

- Frequency of signal: $f = 250 \text{ Hz}$

$$T = \frac{1}{f} = \frac{1}{250} = 4.00 \times 10^{-3} \text{ s} = 4.00 \text{ ms}$$

- From trace: **1 full cycle spans 2.0 divisions**

$$\text{Timebase setting} = \frac{T}{\text{divisions}} = \frac{4.00 \text{ ms}}{2.0} = 2.00 \text{ ms/div}$$

Revision Tips

- RMS voltage** gives the **DC equivalent** power output

$$V_{\text{rms}} = \frac{V_{\text{peak}}}{\sqrt{2}}$$

- Timebase setting** = time per division on x-axis
- Frequency and time are inversely related:

$$f = \frac{1}{T}$$